

Management of Acute Airway Obstruction in a Multi-Trauma Patient using ATLS Protocols

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1. Case Summary & Clinical Presentation

The patient was an adult male brought to the Emergency Department following a high-velocity **Motor Vehicle Accident (MVA)**. The patient was the restrained driver in a head-on collision. Upon arrival, the patient was conscious but agitated (GCS 13, E3, V4, M6).

During the **Primary Survey**, while maintaining manual in-line stabilization of the cervical spine, I identified a clinical red flag: an intermittent, audible **"gurgling" or "snoring" sound** during the patient's inspiratory phase. This "strange sound" indicated a partial airway obstruction despite the patient's ability to vocalize.

The Intervention

Following the **ATLS "Look, Listen, Feel"** algorithm:

- **Assessment:** Performed a visual inspection of the oropharynx with high-intensity lighting.
- **Finding:** Identified a large, fractured tooth fragment lodged in the posterior pharynx, partially obscured by blood.
- **Resolution:** Used suction to clear secretions and retrieved the tooth fragment using forceps under direct visualization.

2. Measuring the Outcome

The "Measuring Outcomes" component is demonstrated by the immediate physiological improvement and the prevention of secondary complications such as aspiration or total occlusion.

Table 1: Clinical Data and Intervention Impact

Clinical Parameter	Initial Assessment (Pre-Intervention)	Post-Intervention (Airway Cleared)	Clinical Significance
Airway Patency	Partial obstruction (Audible gurgling).	Clear and patent; quiet respiration.	Resolved mechanical obstruction.
Oxygen Saturation (SpO2)	93% on 15L	99% on 15L	Improved gas exchange

Respiratory Rate	24 breaths/min (Tachypneic).	18 breaths/min.	Reduced work of breathing.
Mental Status	Agitated; "V4" (Confused).	Calmer; "V5" (Oriented).	Improved cerebral oxygenation.

3. Clinical Reasoning & Protocol Adherence

This case illustrates the critical importance of the **ATLS 11th Edition** Step "A" (Airway).

- **Avoidance of "Distraction Bias":** In multi-trauma, it is common to focus on obvious bleeding or limb deformities. By strictly following the A-B-C sequence, the life-threatening airway issue was resolved before it could escalate.
- **The "Listen" Phase:** The decision to intervene was prompted by an auditory cue (the strange sound) rather than a visual one, proving that "A" is a multi-sensory assessment.

4. Reflection on Practice

What Went Well

The systematic discipline of the ATLS framework prevented me from assuming the airway was safe just because the patient was talking. The prompt removal of the tooth prevented **aspiration pneumonia** or a sudden total occlusion during transport to CT imaging.

Areas for Improvement

During the intervention, I noted a brief delay in accessing the Magill forceps, which were not in the immediate "primary" airway tray. This highlighted a need for better equipment layout in the trauma bay.

Action Plan for Future Practice

- **Equipment Audit:** I have requested a review of our trauma trolley to ensure "Foreign Body Removal" tools (Magill forceps, rigid suction) are at the bedside for all Level 1 Traumas.
- **Documentation:** I will ensure all future trauma notes explicitly document "Airway Sounds" rather than just "Airway Patent" to capture more granular clinical data for future audits.

5. Conclusion

By applying evidence-based ATLS protocols to this case, I was able to measure a direct improvement in the patient's respiratory status (SpO2 increase of 6%). The outcome was a stabilized patient who could safely proceed to definitive care for their associated fractures without the morbidity of an aspirated foreign body.